

How does vegetation differ between urban and rural landscapes? A comparative NDVI analysis using Sentinel-2 imagery

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Software: QGIS, Copernicus Browser

Data Source: Sentinel-2 Satellite Imagery

1. Introduction

Vegetation plays an important role in regulating climate, improving air quality and supporting biodiversity. This study compares vegetation density between Daural(rural) and Kanker Khera(urban) using Sentinel-2 imagery and the Normalised Difference Vegetation Index(NDVI). NDVI was calculated using Red(B04) and Near Infrared Bands(B08), followed by descriptive statistical analysis and NDVI classification into four vegetation density classes. Results showed that Daurala exhibited a higher mean NDVI (0.451) than Kanker Khera(0.237). Land Cover classification further revealed that one-third of Daurala consisted of Dense vegetation compared to only a small proportion of Kanker Khera. The finding demonstrate the usefulness of Remote Sensing for quantifying differences in vegetation distribution between rural and urban landscapes. This finding can also be used to plan plantations drives as the NDVI images shows the parts of the areas where vegetation is low.

2. Aim

The aim of this study was to compare vegetation density between an urban area (Kanker Khera) and a rural area (Daurala) using Sentinel-2 imagery and NDVI analysis.

3. Hypothesis

The hypothesis we make here is that urban areas will have a lower NDVI than rural areas because generally areas with a larger agricultural area have contain a greater actively growing vegetation.

4. Study Area

Kanker Khera is a suburban locality in Meerut characterised by residential development, commercial areas, roads, and scattered green spaces.



Figure 1. True Colour Image of Kanker Khera

Daurala is a predominantly rural settlement located north of Meerut, characterised by extensive agricultural land, particularly sugarcane cultivation, together with villages and irrigation infrastructure.



Figure 1. True Colour Image of Daurala

5. Objectives

- Calculate NDVI for both study areas.
 - Compare descriptive statistics.
 - Classify vegetation density.
 - Compare vegetation distribution.
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6. Data and Methods

Data

- Sentinel-2 imagery
- Copernicus Browser

Software

- QGIS
- Copernicus Browser

Methodology

The following workflow was followed:

1. Cloud-free Sentinel-2 images covering the study areas were selected using the Copernicus Browser for both rural and urban areas.
2. Band 4 (Red) and Band 8 (Near Infrared) were downloaded as GeoTIFF files.
3. Both bands were imported into QGIS.
4. The Raster Calculator was used to calculate NDVI using the standard equation:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

5. The resulting NDVI rasters were then reclassified using the “reclassify by table” tool.
6. Then the “raster layer unique values report” tool was used to estimate the percentages of each type of vegetation in both areas .

4. Statistical Analysis

Descriptive statistics were calculated for the NDVI raster of each study area to quantify vegetation characteristics. The statistics included the mean, minimum, maximum, and standard deviation of NDVI values. The mean NDVI was used to compare the average vegetation density between the urban and rural study areas, while the minimum and maximum values indicated the range of vegetation conditions present. The standard deviation was calculated to assess the variability of vegetation density within each landscape.

To further compare vegetation distribution, the NDVI rasters were reclassified into four vegetation-density classes based on NDVI values:

NDVI Range	Vegetation Class
-1.0 - 0.2	Very Low Vegetation/Built Areas
0.2 - 0.4	Sparse Vegetation
0.4 - 0.6	Moderate Vegetation
0.6 - 1.0	Dense Vegetation

The number of pixels within each class was calculated and converted into percentages of the total study area. This enabled a quantitative comparison of vegetation distribution between Kanker Khera and Daurala.

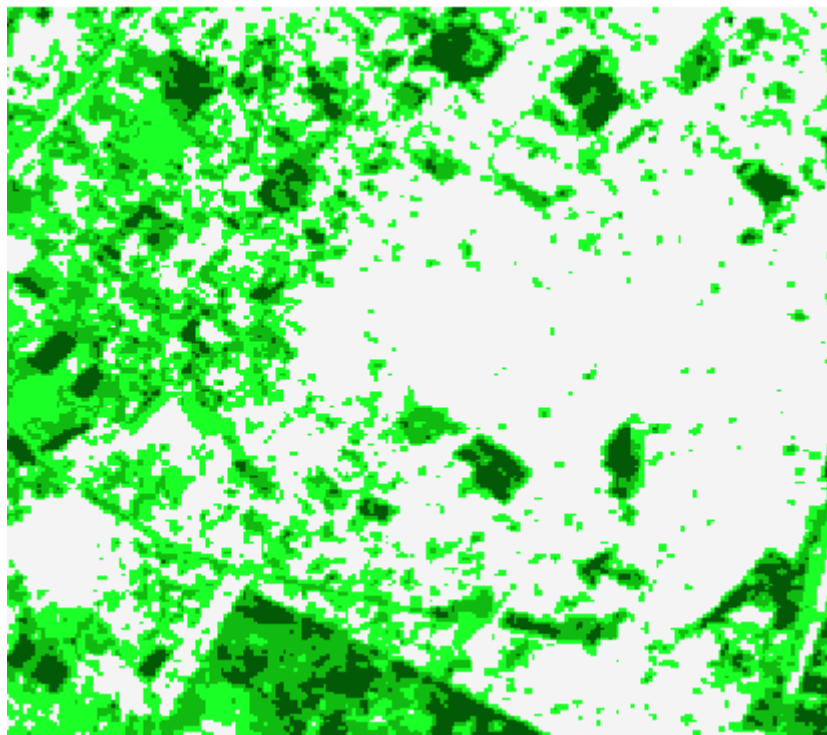


Figure 3. NDVI Image of Kanker Khera

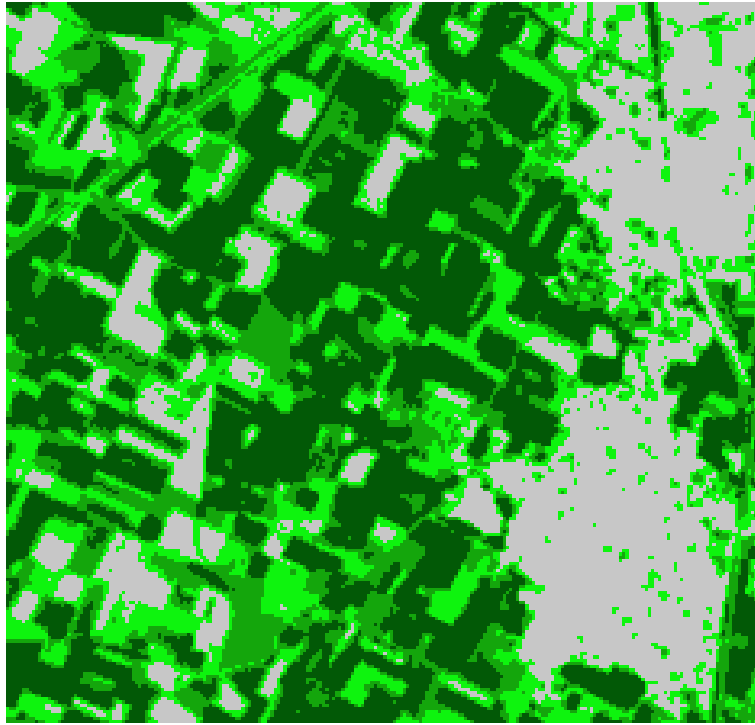


Figure 3. NDVI Image of Daurala

5. Results

5.1 NDVI Statistics

Descriptive statistics showed clear differences in vegetation density between the urban study area (Kanker Khera) and the rural study area (Daurala). Daurala exhibited a substantially higher mean NDVI (0.451) than Kanker Khera (0.237), indicating greater overall vegetation density. Both study areas contained areas of high vegetation, as reflected by similar maximum NDVI values (0.960 and 0.944, respectively). The rural area also exhibited a higher standard deviation, indicating greater variability in NDVI values across the landscape.

Table 1. Descriptive Statistics of NDVI Values

Stastic	Urban(Kanker Khera)	Rural(Daurala)
Mean NDVI	0.237	0.451
Min NDVI	0.053	0.182
Max NDVI	0.944	0.960
Std Dev	0.187	0.269

4.2 NDVI Classification

The classified NDVI maps showed noticeable differences in vegetation distribution between the two study areas.

Kanker Khera was dominated by very low vegetation, with over half of the study area (55.68%) falling within the lowest NDVI class. Dense vegetation occupied only 6.21% of the urban landscape.

In contrast, Daurala contained a much greater proportion of dense vegetation (34.66%), while only 24.02% of the study area was classified as very low vegetation. Moderate vegetation was also more common in Daurala than in Kanker Khera.

Table 2. Percentage of land within each NDVI class for Kanker Khera

NDVI Class	Interpretation	Percentage
1	Very low/Built area	55.78%
2	Sparse	24.59%
3	Moderate	13.43%
4	Dense	6.20%

Table 3. Percentage of land within each NDVI class for Daurala

NDVI Class	Interpretation	Percentage
1	Very low/Built area	24.62%
2	Sparse	20.99%
3	Moderate	19.74%
4	Dense	34.65%

5. Discussion

The results clearly demonstrate that vegetation density differs substantially between the urban area of Kanker Khera and the rural area of Daurala. The higher mean NDVI and greater proportion of moderate and dense vegetation in Daurala support the hypothesis that rural landscapes generally contain more vegetation than urban environments.

The dominance of dense vegetation in Daurala is likely due to the extensive agricultural land surrounding the village. Large areas of sugarcane and other crops produce strong vegetation signals, resulting in higher NDVI values. Although agricultural fields are managed by humans rather than occurring naturally, they still represent healthy and actively growing vegetation that strongly reflects near-infrared radiation.

In contrast, Kanker Khera is characterised by residential buildings, roads, commercial areas, and other impervious surfaces. These land-cover types contain little or no vegetation and therefore produce lower NDVI values. Vegetation within the urban area was mainly confined to scattered trees, gardens, parks, and small green spaces, resulting in a much larger proportion of very low vegetation.

Although both study areas recorded similar maximum NDVI values, these high values represented only small portions of the urban landscape, whereas dense vegetation occupied a much larger proportion of the rural study area. This demonstrates that individual patches of healthy vegetation can exist within urban environments, but they are insufficient to significantly increase the overall vegetation density of the landscape.

This study also demonstrates the usefulness of remote sensing and NDVI as efficient tools for comparing vegetation patterns across different landscapes. Satellite imagery allows vegetation to be assessed over large areas without the need for extensive field surveys, making it valuable for environmental monitoring and land-cover analysis.

However, several limitations should be considered. NDVI measures vegetation greenness rather than vegetation type and cannot distinguish between natural forests, agricultural crops, or managed grassland. The results also represent vegetation conditions on the date the satellite image was acquired and may vary seasonally due to crop cycles, rainfall, and weather conditions. Future studies could analyse multiple dates throughout the year to investigate seasonal changes in vegetation cover.

7. Conclusion

This study successfully compared vegetation density between the urban area of Kanker Khera and the rural area of Daurala using Sentinel-2 satellite imagery and NDVI analysis within QGIS. Both descriptive statistics and vegetation classification consistently showed that Daurala contained substantially greater vegetation density than Kanker Khera.

The results demonstrated that more than half of Kanker Khera consisted of very low vegetation, while over one-third of Daurala was classified as dense vegetation. These findings support the hypothesis that rural landscapes maintain considerably greater vegetation cover than urban

environments, largely because of extensive agricultural land and the lower proportion of built-up surfaces.

Beyond the findings themselves, this project demonstrates how GIS and remote sensing techniques can be used to quantify vegetation patterns and compare landscapes objectively. The methods used in this study provide a foundation for more advanced environmental investigations, including urban heat island analysis, land-use change assessment, and urban greening strategies.
